§7 Collapse, blockage and collapse path sets — part 1 of 5 (opening, blockage, descriptive state bands)

Source: `paper/section7_collapse.tex`, lines 1–1160 (the opening of the section, with its remark on description levels; `subsec:blockage`: the definition of blockage as used here, its relation to SM, required-environment satisfaction, the E_{req} robust viability kernel, well-formed realization and collapse; `subsec:precollapse_states`: the descriptive state bands $\mathrm{Z}_0, \ldots, \mathrm{Z}_5$), at commit e6e12cd. Status: draft. Part 2 (`07_collapse_2.md`, 1161–2347: external recovery and identity, collapse) and later parts follow. §7 defines no assumption. The Japanese manuscript is authoritative; this is a summary, not a translation, and adds no definition, assumption or result.

Opening of §7

`Reference: sec:collapse_cond`

§7 takes up, for a single ACS individual Σ_i, blockage, the descriptive pre-collapse state bands, external recoverability, collapse, and their expression as trajectory records. Its aim is not to merge these into one failure concept but to keep them apart as states at different levels. In particular blockage, non-realization of well-formed realization, and collapse rest on different judgment conditions and are not identified with one another. Blockage is a local state: at one time, under the realized environment, feasible actions are lost. Collapse is not a loss of function at a single time and not blockage itself; it is a state treated as terminal in the present framework (`def:collapse` of §2). §7 adds state bands for describing the surroundings of collapse and path sets that record the reaching of blockage or collapse. It does not build a general cause theory of collapse, a probabilistic reaching theory, or a theory of distance to collapse. Phase 1 treats a single ACS individual here. Symbols already introduced in earlier sections ($A_{i,t}$, $\mathrm{WF}_{i,t}$, Desc_t, Q) are referred to but not redefined.

Levels of description in §7 (`rem:s7_principle_levels`). Four levels are distinguished. (1) *Principles*: SM, EU and FR are constraint axes for a persisting ACS (§3); §7 does not say that a principle itself holds or fails. (2) *Time-indexed satisfaction predicates*: these carry the truth values. The predicate of SM is true when $x_t \in K$ on its domain of application; that of EU is defined on its domain of application; the local-FR predicate is defined only on $\mathcal{D}_{\mathrm{FR}}$. "False" means defined and false; times at which a predicate is undefined are not counted as times at which it is false. (3) *Functions*: the basic and base functions of §2, the structural functions of §3 and the standard functional forms of TIPA, PSL and ISL of §4; operation, stoppage and decline are stated at this level, distinct from the truth of a predicate. (4) *Diagnostic quantities*: classification C quantities such as the functional axis, health, the descriptive state bands and $\mu_{\mathrm{eff},t}$. This paper does not state, as a general consequence, that a predicate is true because functions operate, nor that a particular function has stopped because a predicate is false; nor does it place a dedicated function for each principle.

The section proceeds as follows: blockage (`subsec:blockage`); descriptive state bands (`subsec:precollapse_states`); external recoverability and identity (`subsec:external_recovery`); collapse (`subsec:collapse_def`); blockage and collapse path sets (`subsec:blk_path_set`, `subsec:col_path_set`); inflow from internal dynamics and relations (`subsec:inflow_paths`); and the boundary to the organizational ACS (`subsec:to_section8`).

Blockage

`Reference: subsec:blockage`

Reference to the definition (`def:blockage_sec2` of §2). Blockage is the emptiness of the feasible action set. §7 does not redefine it and uses it in the form $A_{i,t} = A(x_t, e_t) = \{u \in U \mid f(x_t, u, e_t) \in K\} = \emptyset$ for the internal state $x_t \in X$ and the realized external environment state $e_t \in E_{\mathrm{ext}}$.

Rewriting by the persistence evaluation function (`rem:blockage_pi_form`). Since $K = \{x \in X \mid \Pi(x) \geq 0\}$, $A_{i,t} = \emptyset$ if and only if $\forall u \in U : \Pi(f(x_t, u, e_t)) < 0$.

Type of the environment state (`rem:blockage_environment_type`). The e_t in the definition is an element of the external environment state space E_{ext}, not of the third component E of the 7-tuple; $E \subseteq \{0,1\}^{X \times E_{\text{ext}}}$ is the set of required environmental factors, not a set of environment states. The type of f is $X \times U \times E_{\text{ext}} \to X$, and blockage is determined relative to the realized e_t.

Blockage does not mean "no transition" (`rem:blockage_not_no_transition`). $A_{i,t} = \emptyset$ means that, under e_t, no action keeps the next state in K. It does not mean that $f(x_t, u, e_t)$ is undefined, that physical change stops, or that there is no next time. Blockage is the non-existence of an action that retains feasibility one step ahead, not the non-existence of a transition.

Blockage versus SM, required-environment satisfaction and well-formed realization

SM and blockage are different single-time conditions (`rem:blockage_vs_sm`). The SM predicate at t is the feasible-state condition $\Pi(x_t) \geq 0 \iff x_t \in K$ on the current state. Blockage asks, at the same time and for (x_t, e_t), whether an action exists that keeps the state after one step in K. This paper does not assume, in general, either $x_t \in K \Rightarrow A_{i,t} \neq \emptyset$ or $A_{i,t} \neq \emptyset \Rightarrow x_t \in K$; the truth of the SM predicate and the holding of blockage are not identified.

Blockage and required-environment satisfaction differ (`rem:blockage_vs_required_environment`). The required-environment satisfaction set is $E_{\text{req}}(x_t) = \{e \in E_{\text{ext}} \mid \forall \eta \in E : \eta(x_t, e) = 1\}$. Blockage concerns the realized e_t and is the behavioral-feasibility condition $A(x_t, e_t) = \emptyset$. Hence $e_t \in E_{\text{req}}(x_t)$ is not a defining condition of blockage, and $E_{\text{req}}(x_t) = \emptyset$ is not the same as blockage. An implication between the two requires additional conditions on state transition and feasibility.

The relation of blockage to well-formed realization at the same time follows `rem:blockage_not_collapse_sec2` of §2.

Time-locality of blockage

Blockage is not defined as an absorbing state (`rem:blockage_time_dependence`). $A_{i,t}$ is determined relative to x_t and e_t, so blockage is judged time by time. Even if $A_{i,t} = \emptyset$ holds, $A_{i,s}$ for $s > t$ is determined separately by x_s, e_s and, where needed, the external actions of later sections. This paper does not state $A_{i,t} = \emptyset \Rightarrow A_{i,s} = \emptyset$ for all $s > t$ as a general consequence.

Blockage and the E_{req} robust viability kernel

§3 introduced the E_{req} robust viability kernel $\text{Viab}_f(K)$ to capture states from which one common action preserves feasibility for the state-dependent set $E_{\text{req}}(x)$. Blockage refers to *one realized* environment state e_t, whereas $\text{Viab}_f(K)$ demands robust invariance over *all* environment states of $E_{\text{req}}(x)$. To connect them one needs at least that the realized environment belongs to the required-environment satisfaction set.

Proposition (`prop:blockage_viability_kernel`): blockage and the E_{req} robust viability kernel. If at time $t \in \mathbb{N}_0$ both $e_t \in E_{\text{req}}(x_t)$ and $A_{i,t} = \emptyset$ hold, then $x_t \notin \text{Viab}_f(K)$.

Proof (outline). By contradiction. Suppose $x_t \in \text{Viab}_f(K) = \bigcup_{K' \in \mathfrak{K}} K'$ (`def:viab_core`), where $\mathfrak{K}$ is the family of E_{req}-invariant subsets of K_{req}. Then $x_t \in K'$ for some $K' \in \mathfrak{K}$, and by the E_{req}-invariance of K' there is one action $u^\star \in U$ with $f(x_t, u^\star, e) \in K'$ for all $e \in E_{\text{req}}(x_t)$. Since $e_t \in E_{\text{req}}(x_t)$, $f(x_t, u^\star, e_t) \in K' \subseteq K_{\text{req}} \subseteq K$. By `def:at`, $u^\star \in A_{i,t}$, so $A_{i,t} \neq \emptyset$, contradicting $A_{i,t} = \emptyset$. □

When the realized environment lies outside $E_{\text{req}}(x_t)$ (`rem:blockage_viab_requires_ereq`). The premise $e_t \in E_{\text{req}}(x_t)$ is essential. If $e_t \notin E_{\text{req}}(x_t)$, blockage alone does not in general give $x_t \notin \text{Viab}_f(K)$, because the invariance condition of the kernel quantifies only over environment states in $E_{\text{req}}(x_t)$. The manuscript gives a small example (at the level

of types): $X = \{0,1\}$, $\Pi(x) = x - 1$ (so $K = \{1\}$), $U = \{u\}$, $E_{\text{ext}} = \{e^1, e^2\}$, $E = \{\eta\}$ with $\eta(x,e) = 1 \iff e = e^1$, and $f(1,u,e^1) = 1$ with $f = 0$ for all other arguments. Then $\{1\}$ is E_{req}-invariant, so $1 \in \text{Viab}_f(K)$, yet $A_{i,t} = \emptyset$ at $x_t = 1$, $e_t = e^2$. The proposition is therefore used only as the conditional implication "blockage plus realized required-environment satisfaction implies being outside the kernel".

The converse is not adopted (`rem:viability_kernel_not_blockage`). The converse of the proposition, $x_t \notin \text{Viab}_f(K) \Rightarrow A_{i,t} = \emptyset$, is not adopted as a Phase 1 general claim. Being outside the kernel means being outside the set of states with robust invariance for $E_{\text{req}}(x)$; this paper does not state from that, as a general consequence, that no action leads into K in one step under e_t. A state outside the kernel can have $A_{i,t} \neq \emptyset$. Being outside the kernel is not identified with "heading toward blockage" or with "inflow into the blockage path set"; stating such a dynamical connection requires additional conditions.

Blockage versus collapse

The distinction follows `rem:blockage_not_collapse_sec2` of §2. That a blocked state can move to a state with feasible actions at a later time follows `rem:blockage_time_dependence`; its external recoverability is treated in `subsec:external_recovery`.

Four conditions kept apart (`rem:blockage_fourway_distinction`). §7 distinguishes: $\Pi(x_t) \geq 0$ (SM predicate, feasibility of the current state); $A_{i,t} = \emptyset$ (blockage, loss of one-step feasible actions); $\text{WF}_{i,t} = 0$ (non-realization of well-formed realization); and $\text{Col}_{i,t} = 1$ (collapse). They concern different formal objects, and this paper places no equivalence among them without additional assumptions. In particular it does not state, as a general consequence, that blockage gives $\text{WF}_{i,t} = 0$ or $\text{Col}_{i,t} = 1$. The detailed distinction of blockage and collapse, and the reading of non-recoverability (SC-2 of `def:collapse`) from the collapse side, is deferred to `subsec:collapse_def`.

Range of use of the word "blockage" (`rem:blockage_terminology`). Only the state in which $A_{i,t} = \emptyset$ holds is called blockage. The following are not called blockage for that reason alone: the SM predicate being false; non-satisfaction of required environments; being outside the robust viability kernel; non-realization of well-formed realization; total functional stoppage; collapse. The reading of blockage by maintenance phase and recovery phase, its relativity to the one-step transition specification, and its relation to the everyday word "deadlock" follow `rem:blockage_scope_sec2` of §2.

What the definition of blockage does not contain (`rem:blockage_in_precollapse`). A deterministic transition from blockage to collapse, persistence of blockage, recovery from blockage, and a dynamical inflow condition into the blockage path set $\mathfrak{P}_{\text{blk}}$ are not contained in `def:blockage_sec2`, and this paper does not state them as general consequences of that definition alone.

Descriptive state bands around collapse

Reference: `subsec:precollapse_states`

The subsection introduces auxiliary state bands $\text{Z}_0, \ldots, \text{Z}_5$ to describe loss of function and stoppage between blockage and collapse. They are *not* a partition of the formal state space X but classification-C descriptive labels, used so as not to confuse blockage, decline and collapse. Z_0 is the reference band of normal operation; $\text{Z}_1, \ldots, \text{Z}_4$ are the *descriptive pre-collapse state bands*; Z_5 is the terminal-side band corresponding to the collapse state. The range called descriptive pre-collapse state bands is $\text{Z}_1, \ldots, \text{Z}_4$, and Z_0 is not included.

Relation to functional operation and health quantities

Reference to functional operation (`rem:functional_axis_reference`). "Partial malfunction" and "total functional stoppage" in this subsection are *descriptive names* for the operating state of the standard functional forms expressed by TIPA, PSL and ISL of Σ_i. Their formal structure and same-time operation description are left to §4. The subsection introduces no new functional axis, operation indicator, threshold or total-stoppage judgment map; the concrete operationalization follows the functional axis of §4 (`subsec:functional_axis`). The description does not require that a particular biological organ, neural circuit, hardware module or algorithm has stopped uniformly.

Health quantities and bands are not identified (`rem:health_not_zone_identity`). The PSL- and ISL-related health quantities $h_{\mathrm{psl},t}$, $h_{\mathrm{mem},t}$, $h_{\mathrm{sens},t}$, $h_{\mathrm{eff},t}$ of §5 are classification-C quantities for describing, diagnosing and comparing change between times. Phase 1 places no universal equivalence between a component $h_{q,t}$ being at or below a threshold and classification into a particular band Z_j. That health values alone give blockage or collapse is not stated as a general consequence (`rem:s5_health_noncollapse` of §5). Conversely, it is not assumed that a particular health component equals 0 whenever blockage or collapse holds. Health quantities and bands may be related but are separate objects of description.

The bands

Descriptive bands around blockage and collapse (`rem:descriptive_zones`). The labels $\mathrm{Z}_0, \dots, \mathrm{Z}_5$ are classification-C labels; the ACS is not required to hold or compute a variable Z_k. The index of Z_k does not express temporal order, causal order, a strict scale of severity, a distance or a transition probability.

- (a) Z_0, normal operation, reference band. A reference band on the normal side for comparing states around blockage and collapse; typically main functions are maintained and neither blockage nor marked functional decline is in the foreground. No universal health threshold, T-STAB threshold or threshold on the time-indexed satisfaction predicates is fixed. Since it means no disorder, it is *not* included in the pre-collapse bands.

- (b) Z_1, band of manifest mild disorder. Signs such as mild functional decline, chronic load, increased maintenance cost or declining health are descriptively manifest relative to normal operation. The name does not universally mean that a particular health quantity is below a threshold, and this paper does not state, from being in Z_1, that a satisfaction predicate of SM, EU or FR is false, or blockage, non-realization of well-formed realization or collapse, as a general consequence.

- (c) Z_2, band of partial malfunction. Decline or stoppage of part of the standard functions expressed by TIPA, PSL and ISL is described to a degree distinguishable from normal operation. Even with partial malfunction the structural self of the same individual is not taken to be lost at once; nor is $A_{i,t} = \emptyset$, $\mathrm{WF}_{i,t} = 0$ or $\mathrm{Col}_{i,t} = 1$ stated from Z_2 alone as a general consequence.

- (d) Z_3, band of total functional stoppage. The operating functions relating to TIPA, PSL and ISL are described as stopped as a whole. This state is not newly set up as a formal Phase 1 definition. Total functional stoppage is not the same as blockage, and this paper does not state non-realization of well-formed realization or collapse from it alone as a general consequence.

- (e) Z_4, blockage band. It corresponds to the holding of blockage $A_{i,t} = \emptyset$ of `def:blockage_sec2`, and among the bands is the one whose strict formal condition was fixed before this subsection. Blockage is not identical to the SM predicate being false, non-satisfaction of required environments, being outside the robust viability kernel, non-realization of well-formed realization or collapse. A time described as Z_4 can be descriptively divided into (B1) and (B2) by the dichotomy of blockage of §3 (`prop:blockage_dichotomy`); this is a diagnostic division, not a classification of causes (`rem:blockage_dichotomy`).

- (f) Z_5, collapse band. It corresponds to the terminal-side state in which the collapse indicator $\mathrm{Col}_{i,t} = 1$ of `def:collapse` holds. The formal definition and absorbing character of collapse are given in §2 (`def:collapse`, `prop:collapse_basic`), and the scope of non-recoverability in the present framework is treated in `subsec:collapse_def`. The defining condition of Z_5 is not anticipated or added here.

Logical relations among the bands

The bands are not a partition of the state space (`rem:Z_not_partition`). $\mathrm{Z}_0, \dots, \mathrm{Z}_5$ are not formally defined as a family of subsets of X, and this paper does not assume that the classification gives a cover or an exclusive partition of X. In particular Z_1, Z_2, Z_3 are descriptive functional-state labels, so cases where they are described at the same time as blockage Z_4 are not excluded. Z_5 corresponds to the collapse terminal of `def:collapse`, and its simultaneous treatment with ordinary ACS state quantities follows the attribution provision of that definition and the convention for trajectory records (`subsec:blk_path_set`).

The bands are not totally ordered (`rem:Z_not_total_order`). The indices $0, \dots, 5$ are organizing numbers and define no strict severity scale. No total order of severity along the indices and no deterministic transition from Z_k to Z_{k+1} is assumed. For example, a move from partial malfunction Z_2 to blockage Z_4, a recovery from total stoppage Z_3 to the normal side, or reaching collapse without passing through blockage or total stoppage is neither affirmed nor denied by the numbering convention alone. This is a separate matter from the definitions of the path sets $\mathfrak{P}_{\mathrm{blk}}$ and $\mathfrak{P}_{\mathrm{col}}$.

Total stoppage and blockage are not identical (`rem:full_stop_not_blockage`). Total functional stoppage Z_3 is a descriptive state about stoppage of the operation of functions relating to TIPA, PSL and ISL; blockage Z_4 is the strict behavioral-feasibility condition $A_{i,t} = \emptyset$ under the realized e_t. This paper does not state $\mathrm{Z}_3 \Rightarrow \mathrm{Z}_4$ or the converse as a general consequence. Both may be described at the same time, but the definition does not require their simultaneous holding.

Distinction from the satisfaction predicates, well-formed realization and collapse

Bands are not identified with a satisfaction predicate being false (`rem:zones_not_principle_failure`). $\mathrm{Z}_1, \mathrm{Z}_2, \mathrm{Z}_3$ are descriptive labels for decline or stoppage at the function level (`rem:s7_principle_levels`). From any of them being described, it is not stated that some SM, EU or FR predicate is false; conversely, from a predicate being false, classification into a particular $\mathrm{Z}_1, \mathrm{Z}_2, \mathrm{Z}_3$ is not assumed. Blockage Z_4 is likewise not identical to the SM predicate being false. Thus the truth values of the satisfaction predicates are distinct from the description of Z bands.

Pre-collapse bands and non-realization of well-formed realization (`rem:precollapse_not_wf_failure`). This paper does not state, from a description in any of $\mathrm{Z}_1, \dots, \mathrm{Z}_4$ alone, $\mathrm{WF}_{i,t} = 0$ as a general consequence. Even if partial malfunction, total stoppage or blockage exists, the formal realization of the structural self of the same individual need not be lost at that time. Conversely, from $\mathrm{WF}_{i,t} = 0$ it is not stated which band of this subsection the state belongs to. The formal relation between well-formed realization and collapse is treated with limited scope in `subsec:collapse_def`.

The term “pre-collapse state bands”

Usage (`rem:precollapse_terminology`). $\mathrm{Z}_1, \mathrm{Z}_2, \mathrm{Z}_3, \mathrm{Z}_4$ are collectively called the *descriptive pre-collapse state bands*, abbreviated *pre-collapse state bands*. “Pre-collapse state” without “band” is a general explanatory phrase that intends no classification into a particular band. Z_0 is not included: it is a comparison reference on the normal side and means neither disorder nor a pre-collapse state. “Pre-collapse” does not mean that these states appear as precursor stages of collapse in time or cause. This paper does not state, from being in Z_j ($j = 1, 2, 3, 4$), $\exists s > t : \mathrm{Col}_{i,s} = 1$ as a general consequence, and it does not make prior passage through any of $\mathrm{Z}_1, \dots, \mathrm{Z}_4$ a requirement for collapse. The name has only the classificatory sense: bands that describe disorder and blockage states around collapse while being distinct from collapse.

Bands, recovery, collapse and path sets (`rem:zones_no_path_implication`). This paper does not state, from being described in a particular $\mathrm{Z}_1, \ldots, \mathrm{Z}_4$ alone, external recoverability, membership in the collapse-reaching path set, or the reaching of collapse, as a general consequence. For the blockage path set: when a time described as Z_4 belongs to the ACS-record-valid time domain $I_{\mathrm{ACS}}(\sigma)$ (`def:acs_valid_time_domain`) of a trajectory record σ (`def:trajectory_space`), then $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ by `def:blk_path_set`. From being described in Z_1 to Z_3 alone, this membership is not stated as a general consequence. The formal conditions for external recoverability, the blockage path set and the collapse-reaching path set are given in `subsec:external_recovery`, `subsec:blk_path_set` and `subsec:col_path_set`.

Check status. This summary adds no new definition, assumption or result. The example of `rem:blockage_viab_requires_ereq` is the manuscript's own. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.